

Structural Consciousness

Agenthood, Cognition, and the Constitution of a Subjective Field

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Abstract

Structural Consciousness develops an **organization-level constitutive account** of the causal organization by which the active processes of a physical cognitive system are proposed to constitute a unified, structured, agent-relative subjective experience from one continuing point of view. The construction proceeds from agenthood, through cognition with retained historical organization, to a conjunctive active-field organization. The resulting organization is termed a **subjective field**: it is differentiated, causally integrated, effective at the agent scale, capacity-coupled, persistently self-indexed, situated relative to self, interface, and world, evaluatively sensitive to viable continuation, and temporally unified. The definition is substrate-independent, while each realization remains physically specific.

The content formalism grounds qualitative occurrence in realization-level differentiation, distinguishes novel occurrences from recurrent qualitative types, and permits partial reinstatement of historically formed sensory, bodily, and evaluative relations during later perception and recall. Viable-reserve relations enter the field through field-integrated valence and motivation. Definition 8 specifies the technical category of structural consciousness; the account identifies complete realization of that organization with the organization of the target subjective experience. Positive instantiation requires realization-specific causal evidence for the complete constitutive organization, while external description of that organization and the agent's first-person standpoint remain distinct epistemic relations.

Introduction

Consciousness research addresses distinct explanatory targets including conscious access, conscious level, qualitative character, phenomenal structure, and unity.[38][39][43][44] Structural Consciousness (SC) addresses a constitutive question: **what causal-organizational structure would a physical cognitive system have to realize for its active processes to constitute a unified subjective experience from one continuing point of view?** Phenomenal-unity research concerns how distinct experiences belong together within one subject,[40][41] while first-person subjectivity and third-person scientific description remain distinct methodological relations.[47][48]

SC specifies the constitutive organization of the target subjective field; conscious level, wakefulness, conscious access, and behavioural responsiveness remain distinct assessment targets.

SC uses an **organization-level constitutive analysis** with three dependency stages. Agenthood specifies a continuing bearer through identity-preserving continuation, an effective boundary, bidirectional causal exchange, and viability under a declared disturbance class. Cognition adds retained historical change, future causal efficacy of retained organization, composite formation and recruitment, endogenous propagation, competition among incompatible continuations, and recursive modification of later candidate availability. Structural consciousness then requires the active relational field to be differentiated, causally integrated, agent-effective, capacity-coupled, persistently self-indexed, situated relative to self, interface, and world, evaluatively sensitive to viable continuation, and temporally unified. The conjunction is non-compensatory.

Behavioural resemblance, constitutive-organizational realization, and first-person epistemic access are distinct claims.[1][2][3][43][44] Let $B(A)$ denote behavioural resemblance, $S(A)$ realization under SC, and $\mathcal{T}_{SC}$ the definitions and premises of the account. $\mathcal{T}_{SC}$ contains no rule identifying $B(A)$ with $S(A)$; behavioural

resemblance is therefore not a sufficient SC realization criterion. This distinction is especially relevant to artificial systems whose verbal behaviour may be generated by architectures with different persistence and causal organization.[4][5]

For a physical system Y , an **admissible realization interpretation** ρ_Y assigns realization-specific states, relations, interventions, identity relations, and capacity variables to the theory with causal fidelity. It induces a structural model $\mathfrak{M}_Y^{\rho_Y}$, and at fixed ρ_Y and analysis context χ ,

$$\boxed{\mathfrak{M}_Y^{\rho_Y} \models \text{SC}_\chi(Y, t)}$$

is the technical membership criterion. The constitutive hypothesis identifies complete physical realization of this category with the organization of the target unified, structured, agent-relative subjective experience. A relation is **constitutive in the present model** when it belongs to that organization rather than functioning only as an upstream cause, downstream consequence, behavioural correlate, or enabling condition. Empirical realization assessment and comparative evaluation of the constitutive hypothesis remain distinct. Putative minimally structured or non-agentic forms of phenomenal experience discussed elsewhere are not classified by the present agent-level construction.[42]

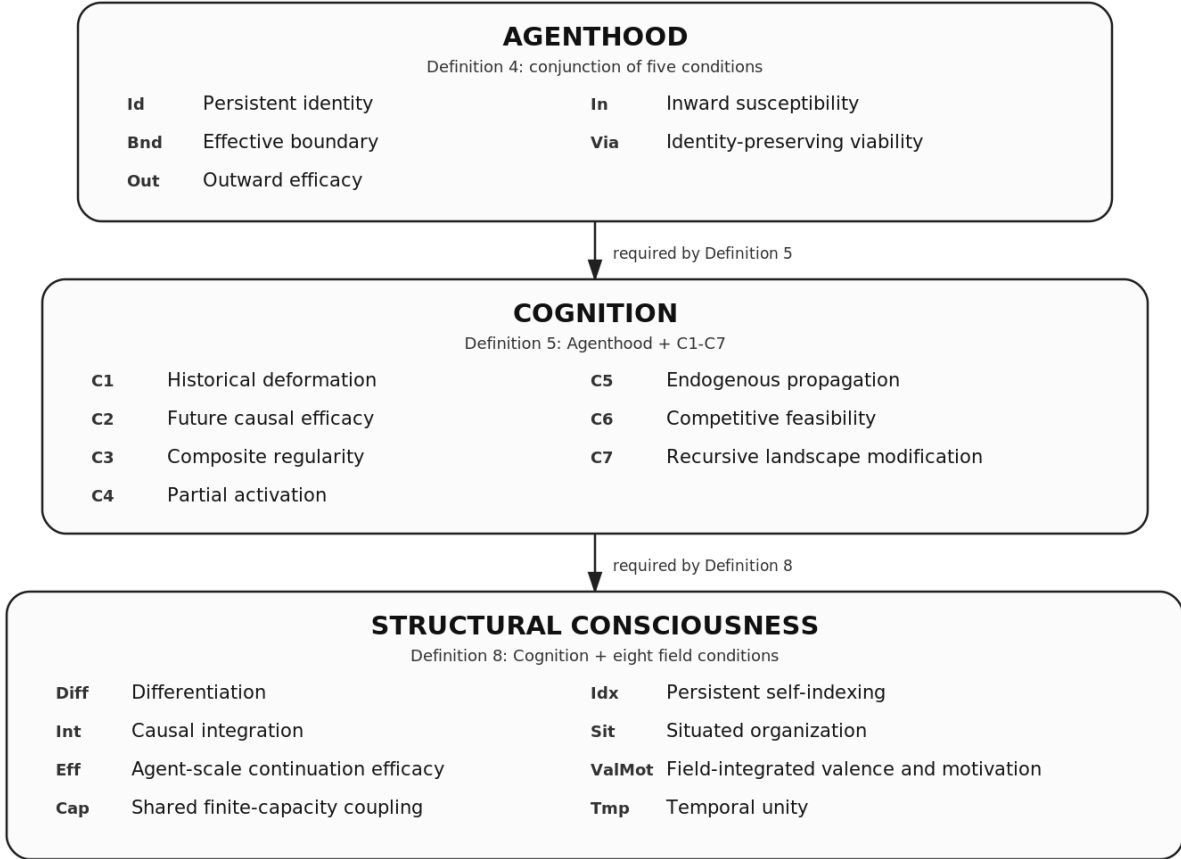


Figure 1: Logical hierarchy of Agenthood, Cognition, and Structural Consciousness, showing five agenthood conditions, seven cognition conditions, and eight field conditions.

Figure 1. Logical construction of Agenthood, Cognition, and Structural Consciousness. Agenthood contains the five conditions of Definition 4; cognition adds C1–C7 of Definition 5; structural consciousness adds the eight field conditions of Definition 8.

1. Agenthood: context, identity, and viability

Because the target concerns subjective experience from one continuing point of view, the bearer of the analysis cannot be left implicit. The first stage identifies a causally coherent candidate system at a selected scale, with a boundary, a temporal horizon, and conditions under which its identity persists.[19][20]

1.1 Analysis context and feasible continuation

Definition 1 (analysis context). Fix

$$\chi = (s, h, \mathcal{D}_A),$$

where s is the selected causal scale, $h > 0$ is the causal horizon, and $\mathcal{D}_A$ is a nontrivial disturbance class under which viability is assessed. The corresponding agent description includes the admissible state space $\mathcal{X}_A(t)$, actual state $x_A(t)$, identity-continuation family κ , effective boundary $\partial A(t)$, inward and outward exchange $J_A(t)$, and admissible transition relations over $[t, t + h]$.

Definition 2 (identity continuation). For system tokens A, B and $t' \geq t$, let

$$\kappa_{A,B}^{t,t'} \subseteq \mathcal{X}_A(t) \times \mathcal{X}_B(t')$$

and interpret $\kappa_{A,B}^{t,t'}(x, x')$ as saying that x' is an identity-preserving continuation of x under the declared realization interpretation and selected scale. For same-system continuation write $\kappa_A^{t,t'} := \kappa_{A,A}^{t,t'}$. The family is reflexive at a fixed time and temporally compatible:

$$\begin{aligned} &\kappa_{A,A}^{t,t}(x, x), \\ &\kappa_{A,B}^{t,t'}(x, x') \wedge \kappa_{B,C}^{t',t''}(x', x'') \implies \kappa_{A,C}^{t,t''}(x, x''). \end{aligned}$$

The identity-continuation relation is a realization-level criterion. In a continuous realization, it is instantiated through uninterrupted continuation of the realization-declared identity-supporting physical organization at the selected scale; persistence of particular material tokens is not required. The account additionally assumes that an actual agent identity occupies one singular realized continuation of its latest identity-bearing organization. This singular-continuation hypothesis is not derived from Definition 8 or from general physics and is not, without further premises, a theorem of numerical personal identity.[29][30][31]

For an initial state x at time t , let $I_A[t, t + h]$ be the realization-relevant ordered index set and let $\mathcal{H}_A(x, t; h)$ be the admissible state histories x_γ on that index set. Define the endpoint projection

$$\boxed{\mathcal{R}_A(x, t; h) = \{x_\gamma(t + h) \mid x_\gamma \in \mathcal{H}_A(x, t; h)\}.$$

Definition 3 (feasible continuation set). $\mathcal{R}_A(x, t; h)$ is the set of states reachable at $t + h$ through admissible transition chains. A superscript on $\mathcal{H}_A$ or $\mathcal{R}_A$ denotes the same construction under a declared disturbance, intervention, or schedule. The horizon is nontrivial only when it spans at least two causally dependent transitions such that an admissible modification at an earlier transition can change later continuation structure within the same horizon. The baseline and modified histories used for that claim must be matched before the modification and outside its causal consequences. The disturbance class must likewise contain at least one physically admissible disturbance that changes the candidate's admissible histories or endpoint projection.

The boundary need not coincide with a biological body. A software agent may have a distributed boundary involving processes, memory stores, sensors, tool permissions, output channels, and physical devices. Boundary membership is determined by the causal relations of the declared realization interpretation.

1.2 Agenthood

Five conditions are jointly required at the selected context:

1. **Persistent identity**: at least one feasible later state is related to the present state by κ_A .
2. **Effective boundary**: the declared internal and external variables form a causally meaningful distinction whose cross-boundary paths use the declared exchange structure.
3. **Outward efficacy**: an admissible internal intervention can alter later outward exchange.
4. **Inward susceptibility**: an admissible external intervention can alter the internal state history.
5. **Identity-preserving viability**: for every disturbance in $\mathcal{D}_A$, at least one admissible continuation preserves the agent's identity over the stated horizon.

Let $\mathcal{J}_A^{\text{int}}$ be the declared class of admissible interventions on internal variables that do not directly overwrite the outward-exchange variables used as outcomes, and let $\mathcal{J}_A^{\text{ext}}$ be the corresponding class for external conditions that do not directly overwrite the internal variables used as outcomes. Let $\mathbf{J}_A^{\text{out}}[\mathcal{H}]$ denote the outward-exchange trajectories induced by a set of histories and $\Pi_A^{\text{int},\mathcal{H}}[\mathcal{H}]$ their projection onto the declared internal variables. Then

$$\begin{aligned}\text{Id}_{A,\chi}(t) &\iff \exists x' \in \mathcal{R}_A(x_A(t), t; h) : \kappa_A^{t,t+h}(x_A(t), x'), \\ \text{Out}_{A,\chi}(t) &\iff \exists \iota_{\text{int}} \in \mathcal{J}_A^{\text{int}} : \mathbf{J}_A^{\text{out}}[\mathcal{H}_A^{\iota_{\text{int}}}] \neq \mathbf{J}_A^{\text{out}}[\mathcal{H}_A],\end{aligned}$$

and

$$\text{In}_{A,\chi}(t) \iff \exists \iota_{\text{ext}} \in \mathcal{J}_A^{\text{ext}} : \Pi_A^{\text{int},\mathcal{H}}[\mathcal{H}_A^{\iota_{\text{ext}}}] \neq \Pi_A^{\text{int},\mathcal{H}}[\mathcal{H}_A].$$

In these expressions the history sets have the common initial state $x_A(t)$, time t , and horizon h . Using history projections permits inward susceptibility to be identified when a transient internal trajectory changes even if the compared endpoints later converge.

Let $\mathcal{U}_A^{\text{int}}(t; h)$ and $\mathcal{U}_A^{\text{ext}}(t; h)$ be the nonempty, disjoint internal and external variable classes declared at scale s . Let $\mathcal{P}_A^{\partial}(t; h)$ denote the complete represented set of directed causal paths in the assessment model that cross between those two classes in either direction; the realization interpretation may not omit represented bypass paths in order to satisfy the boundary predicate. Let $\text{Uses}_{J_A}(\gamma)$ mean that path γ contains at least one relation assigned to the inward or outward exchange structure. The boundary predicate is

$$\boxed{\text{Bnd}_{A,\chi}(t) \iff \mathcal{U}_A^{\text{int}} \neq \emptyset \wedge \mathcal{U}_A^{\text{ext}} \neq \emptyset \wedge \mathcal{U}_A^{\text{int}} \cap \mathcal{U}_A^{\text{ext}} = \emptyset \wedge \mathcal{P}_A^{\partial}(t; h) \neq \emptyset \wedge \forall \gamma \in \mathcal{P}_A^{\partial}(t; h) : \text{Uses}_{J_A}(\gamma).}$$

The viability predicate is

$$\text{Via}_{A,\chi}(t) \iff \forall d \in \mathcal{D}_A \exists x' \in \mathcal{R}_A^d(x_A(t), t; h) : \kappa_A^{t,t+h}(x_A(t), x').$$

The quantifier order requires an identity-preserving admissible continuation for each declared disturbance; it does not require one fixed policy to succeed under all disturbances.

Definition 4 (agenthood).

$$\boxed{\text{Agent}_{\chi}(A, t) \iff \text{Id}_{A,\chi}(t) \wedge \text{Bnd}_{A,\chi}(t) \wedge \text{Out}_{A,\chi}(t) \wedge \text{In}_{A,\chi}(t) \wedge \text{Via}_{A,\chi}(t).}$$

Agenthood is scale-relative. A component may participate in an agent without itself satisfying Definition 4; a collection forms one agent only when the relevant identity, boundary, exchange, and viability relations occur at that scale.

2. Cognition: retained history and endogenous selection

Agenthood supplies persistence, boundary, exchange, and viability. Cognition additionally requires past interaction to alter the actual present organization and later possibilities, while internally available alternatives can recruit, propagate, compete, and modify the organization from which later alternatives arise.

All seven predicates are evaluated relative to the actual state $x_A(t)$. Past-history witnesses terminate before t ; future histories and interventions express causal dispositions of that state over the declared horizon. A realization declares internal relational substructures, projection maps π_u , a proper-substructure relation $\prec$, and

$$\text{Comp}_A(q; u_1, \dots, u_m), \quad m \geq 2,$$

for a maintained composite q containing declared proper parts. Write $u \preceq v$ when $u = v$ or $u \prec v$. Let $\hat{\pi}_q(x) = \pi_q(x)$ when q is present and defined, and $\hat{\pi}_q(x) = \perp_q$ when q is absent; $\perp_q$ may not conceal an undefined projection. For repeated pattern r and past history η , $N_A(r, \eta; t)$ counts occurrences completed before t .

Matched-history comparisons hold exogenous conditions and boundary interactions fixed outside consequences of the altered relation.[59] For baseline history x_γ and proper part $p \preceq x$, $\text{Match}_A(\gamma, p)$ contains histories generated by an admissible intervention on p that does not directly overwrite the candidate composite or later comparison variables. The notation $q \in \mathcal{K}_A(p, x, t; h)$ means that p is present and proper to q , some admissible future history contains q , the complete q is not imposed across the boundary, and altering p changes whether, when, or how q occurs.

Cognition and field stages share an ordered heterogeneous demand domain $(\mathcal{C}_A, \preceq_C)$ and accessible capacity region $\mathcal{B}_A(t; h) \subsetneq \mathcal{C}_A$, downward closed and upper bounded:

$$c \in \mathcal{B}_A(t; h), \ c' \preceq_C c \implies c' \in \mathcal{B}_A(t; h), \quad \exists b_A(t; h) \ \forall c \in \mathcal{B}_A(t; h) : c \preceq_C b_A(t; h).$$

Let $\Sigma_A(S; t, h)$ be structurally and temporally admissible maintenance schedules, $\mathbf{D}_A(S, \sigma; t, h)$ their heterogeneous aggregate demand, and $\mathbf{a}_{A,u}(\sigma; t, h)$ the direct allocation variables assigned to constituent u . Schedule admissibility is distinct from capacity feasibility, which requires demand in $\mathcal{B}_A(t; h)$; no additive scalar resource budget is assumed.[37]

For $i = 1, \dots, 7$, $C_{i,\chi}(A, t)$ holds exactly when the corresponding condition below is satisfied.

Historical deformation (C_1). Let $\mathcal{W}_{1,A}(t)$ contain tuples $(\eta, \bar{\eta}, u, \bar{x}_A(t))$ in which η reaches $x_A(t)$, the initiating boundary interaction ends before t , $\bar{\eta}$ is matched outside that episode, the projections are defined, and

$$\pi_u(x_A(t)) \neq \pi_u(\bar{x}_A(t)).$$

Then

$$\boxed{C_{1,\chi}(A, t) \iff \mathcal{W}_{1,A}(t) \neq \emptyset.}$$

Future causal efficacy (C_2). For a witness in $\mathcal{W}_{1,A}(t)$, let $\mathcal{J}_A^{\text{ret}}(u, \bar{x}_A(t))$ contain admissible interventions replacing the retained projection by its matched-history value without directly overwriting other present comparison variables or later outcomes. Let $\text{Dir}_A(\iota_u)$ denote the realization-level assignments made directly by the intervention. For a declared later substructure v , let $\pi_{v,\iota_u,\tau}^{\text{nd}}$ be a nonempty realization-declared restriction of π_v to later comparison components whose values at τ are not fixed by those direct assignments. These components may represent state, relational, timing, feasibility, or candidate-availability properties within the same physical organization. Define

$$\Pi_{A,v,\tau,\iota_u}^{\text{nd}}[\mathcal{H}] = \left\{ \pi_{v,\iota_u,\tau}^{\text{nd}}(x_\gamma(\tau)) \mid x_\gamma \in \mathcal{H} \right\}.$$

Let $\mathcal{U}_A^{\text{nd}}(\iota_u, \tau)$ contain the declared v for which this restricted projection is defined on every compared state. Future causal efficacy requires a later realized consequence beyond persistence or redescription of what the retained-state intervention directly assigns:

$$\boxed{C_{2,\chi}(A, t) \iff \begin{aligned} &\exists(\eta, \bar{\eta}, u, \bar{x}_A(t)) \in \mathcal{W}_{1,A}(t), \exists \iota_u \in \mathcal{J}_A^{\text{ret}}(u, \bar{x}_A(t)), \\ &\exists h' \in (0, h], \exists \tau \in I_A(t, t+h'), \exists v \in \mathcal{U}_A^{\text{nd}}(\iota_u, \tau) : \\ &\Pi_{A,v,\tau,\iota_u}^{\text{nd}}[\mathcal{H}_A^{\iota_u}(x_A(t), t; h')] \neq \Pi_{A,v,\tau,\iota_u}^{\text{nd}}[\mathcal{H}_A(x_A(t), t; h')]. \end{aligned}}$$

The later consequence may concern recruitment, timing, candidate availability, feasibility, or other realization-specific organization; persistence or redescription of only the directly assigned retained variables does not satisfy C_2 .

Composite regularity (C_3). Repeated relational history must causally contribute to a maintained composite. There are a repeated pattern r , a composite q with at least two declared proper parts, and matched histories $\eta, \bar{\eta}$ such that η reaches the actual state, r occurs at least twice in η and fewer times in $\bar{\eta}$, and the presence or realized projection of q differs. A merely hard-coded q following repeated episodes does not satisfy the condition.

Partial activation (C_4).

$$\boxed{C_{4,\chi}(A, t) \iff \exists p, q : q \in \mathcal{K}_A(p, x_A(t), t; h)}.$$

Endogenous propagation (C_5). Let $\Gamma_A(p, q, t; h)$ denote matched history pairs satisfying the recruitment conditions used in $\mathcal{K}_A$. For witnesses used in C_5 , the relevant occurrence and outward-divergence sets have least elements. Let $\tau_q(\gamma)$ be the first occurrence of q and $\tau_{\text{out}}(\gamma, \bar{\gamma})$ the first outward-exchange divergence, or $+\infty$. Then

$$\boxed{C_{5,\chi}(A, t) \iff \begin{aligned} &\exists p, q, \gamma, \bar{\gamma}, u, \tau : (x_\gamma, x_{\bar{\gamma}}) \in \Gamma_A(p, q, t; h) \wedge u \not\leq p \\ &\wedge t < \tau_q(\gamma) \leq \tau < \tau_{\text{out}}(\gamma, \bar{\gamma}) \wedge x_\gamma(\tau), x_{\bar{\gamma}}(\tau) \in \text{Dom}(\pi_u) \\ &\wedge \pi_u(x_\gamma(\tau)) \neq \pi_u(x_{\bar{\gamma}}(\tau)). \end{aligned}}$$

The same matched recruitment contrast must produce an internal difference outside the initially altered proper part at or after the first occurrence of q , before any outward-exchange divergence.

Competitive feasibility (C_6). At least two distinct candidate completions must each be capacity-feasible while unrestricted joint maintenance is capacity-infeasible:

$$\boxed{C_{6,\chi}(A, t) \iff \begin{aligned} &\exists p, q_1, q_2, \sigma_1, \sigma_2 : q_1, q_2 \in \mathcal{K}_A(p, x_A(t), t; h) \wedge q_1 \neq q_2 \\ &\wedge \sigma_1 \in \Sigma_A(q_1; t, h) \wedge \mathbf{D}_A(q_1, \sigma_1; t, h) \in \mathcal{B}_A(t; h) \\ &\wedge \sigma_2 \in \Sigma_A(q_2; t, h) \wedge \mathbf{D}_A(q_2, \sigma_2; t, h) \in \mathcal{B}_A(t; h) \\ &\wedge \Sigma_A(\{q_1, q_2\}; t, h) \neq \emptyset \\ &\wedge \forall \sigma_{12} \in \Sigma_A(\{q_1, q_2\}; t, h) : \mathbf{D}_A(\{q_1, q_2\}, \sigma_{12}; t, h) \notin \mathcal{B}_A(t; h). \end{aligned}}$$

This is a local candidate-selection condition; field-wide capacity coupling is imposed separately in Section 4.

Recursive landscape modification (C_7). Let $\text{sel}(q_a)$ actualize candidate q_a and $\text{alt}(q_a)$ be a matched alternative completion or non-actualization under the same exogenous conditions. These modifications may establish initial actualization but may not directly overwrite later variables used to compare candidate availability. C_7 requires $t < t' < t + h$, $h' > 0$, $t' + h' \leq t + h$, candidate $q_a \in \mathcal{K}_A(p, x_A(t), t; t' - t)$, and matched later states

$$x_a \in \mathcal{R}_A^{\text{sel}(q_a)}(x_A(t), t; t' - t), \quad x_b \in \mathcal{R}_A^{\text{alt}(q_a)}(x_A(t), t; t' - t),$$

with $p' \preceq x_a \wedge p' \preceq x_b$ and

$$\mathcal{K}_A(p', x_a, t'; h') \neq \mathcal{K}_A(p', x_b, t'; h').$$

Partial relational reinstatement is derived rather than an additional cognition conjunct: a present cue can reinstate an independently declared nonempty subset of historically formed sensory, bodily, world-related, or evaluative relations under a realization-specific correspondence relation fixed independently of the later comparison, without exact state replay. Endogenous reinstatement additionally binds the changed reinstated signature to the same proper-part intervention that witnesses recruitment.

Definition 5 (cognitive agent).

$$\text{Cognitive}_\chi(A, t) \iff \text{Agent}_\chi(A, t) \wedge \bigwedge_{i=1}^7 C_{i,\chi}(A, t).$$

The predicates form a dependency-structured causal organization: retained history becomes causally effective, partial structure recruits maintained composites, recruitment propagates internally, finite capacity constrains alternatives, and actualization changes later candidate availability.

3. Situated organization: self, interface, and world

A situated agent organizes relations that differ in how they bear on its own continuation and boundary. The distinction is functional and causal rather than anatomical. Let $\mathcal{Q}_A(t; h)$ be the comparison domain of active or causally relevant relations. **Definition 6 (relational comparison maps).** Define

$$\mathbf{i}_A : \mathcal{Q}_A(t; h) \rightarrow \mathcal{O}_A^i, \quad \mathbf{c}_A : \mathcal{Q}_A(t; h) \rightarrow \mathcal{O}_A^c, \quad \mathbf{p}_A : \mathcal{Q}_A(t; h) \rightarrow \mathcal{O}_A^p,$$

where the codomains are preordered comparison domains, $\mathbf{i}_A(r)$ orders the effect of alteration of r on identity continuation, $\mathbf{c}_A(r)$ orders the reliability and economy of endogenous controllability, and $\mathbf{p}_A(r)$ orders persistence independently of endogenous control. Let

$$\Phi_A(r) = (\mathbf{i}_A(r), \mathbf{c}_A(r), \mathbf{p}_A(r)).$$

Before assessment, a realization declares classification regions Ω_A^S , Ω_A^Γ , and Ω_A^W . The active relational classes are

$$\Lambda_A^\alpha(t; h) = \{r \in \mathcal{Q}_A(t; h) \mid \Phi_A(r) \in \Omega_A^\alpha\}, \quad \alpha \in \{S, \Gamma, W\}.$$

Λ_A^S contains self-related organization, Λ_A^Γ interface organization mediating exchange across ∂A , and Λ_A^W world-related organization exhibiting partial exogenous persistence. The classes may overlap and need not exhaust the domain; a single relation may therefore witness more than one situated role only when the independently declared classification rules place it in each corresponding region. A tool, prosthesis, or external resource can change classification as control, identity coupling, or exogenous persistence changes.[21][22][23][26][27]

4. The structural-consciousness field

Agenthood and cognition still permit active processes that are mutually disconnected, differently indexed, or unable to belong to one temporally maintained organization. The final stage specifies the relations by which the active contents of one cognitive agent form the single agent-relative organization termed a **subjective field**.

4.1 Field specification and realized differentiation

Let the structural model supply the realized state history x_A^* over the assessment interval, with $x_A^*(t) = x_A(t)$. Define

$$G_A(t; h) \preceq x_A^*[t, t + h]$$

as the active relational substructure of that history, extending the declared realization-level substructure relation to temporally indexed relational histories. Write $V(G_A(t; h))$ for the constituent-track set of that relational structure, and let $M_A(t; h) \subseteq V(G_A(t; h))$ be the nonempty class of **major constituent tracks** over which integration is asserted. A track may change realization-specific state while retaining its declared functional and causal continuity through the interval. Applications declare the tracks before assessment using realization-specific activation, maintenance, and relevance rules.

For $g_i \in M_A(t; h)$, let $\pi_i^{\mathcal{H}}$ project complete histories onto the variables and relations assigned to that track, and let $\Pi_A^{\mathcal{H}}$ project them onto declared whole-agent outcomes over the horizon, such as available action, maintained goals, accessible memory, or other agent-level continuations. Let $\mathcal{I}_i$ be the admissible interventions on the variables assigned to g_i . Such interventions must preserve the declared system boundary, respect realization-specific intervention constraints, and must not directly overwrite the outcome variables used to establish their effects.

In Definition 7 only, the common arguments $(t; h)$ are suppressed on G_A , M_A , the situated relation classes, and $\mathcal{B}_A$ to reduce notation.

Definition 7 (active field).

$$\mathfrak{X}_A^x(t) := (G_A, M_A, \kappa_A, \Lambda_A^S, \Lambda_A^I, \Lambda_A^W, \mathcal{B}_A, \mathcal{H}_A(x_A(t), t; h)).$$

Here *field* denotes a distributed causal organization, not a separate substance or physical field. Definition 7 supplies the realization-level field object on which the field predicates are evaluated. Agenthood specifies the continuing causal system; cognition specifies its retained and endogenously propagating organization; Definition 8 specifies the conditions under which the active contents constitute the target subjective field.

Figure 2. Realization schema for the temporally extended active field. Major constituent tracks are maintained through changing field slices; solid causal and dashed capacity relations represent horizon-level relations with time-respecting witnesses. Self-, interface-, world-related and evaluative relations are internal to the field.

For each active track $g_i \in M_A$, let u_{g_i} denote its realized relational occurrence in the actual field history. A realization-grounded content map $\hat{c}_A$ assigns content identity from declared realization-level state or relational properties, not from labels or track names. Define $c_A(g) = \hat{c}_A(u_g)$. Differentiation requires

$$\text{Diff}_{A,x}(t) \iff \exists g_i, g_j \in M_A(t; h) : g_i \neq g_j \wedge c_A(g_i) \neq c_A(g_j).$$

A declared qualitative comparison domain $\mathcal{Z}_A^Q$ supports recurrence and proximity relations among realized differentiated contents. $\equiv_A^Q$ is an equivalence relation for recurrence of one qualitative type across distinct occurrences, and d_A^Q orders realization-specific qualitative proximity without requiring a metric. A current occurrence is novel when no retained qualitative type available before t is equivalent to it. Repeated occurrence can later stabilize a retained qualitative type through the cognition-stage history relations. The same retained type may be instantiated within different object or event composites.

4.2 Causal integration and agent-scale efficacy

Let $\mathcal{G}_A^{\text{cau}} = (M_A, E_A^{\text{cau}})$ be a directed graph in which (g_i, g_j) is an edge when some $\iota_i \in \mathcal{I}_i$ changes the realized history projection of g_j . Causal integration is ordinary connectivity of the underlying undirected graph:

$$\text{Int}_{A,x}(t) \iff \text{Conn}(\text{Und}(\mathcal{G}_A^{\text{cau}}(t; h))).$$

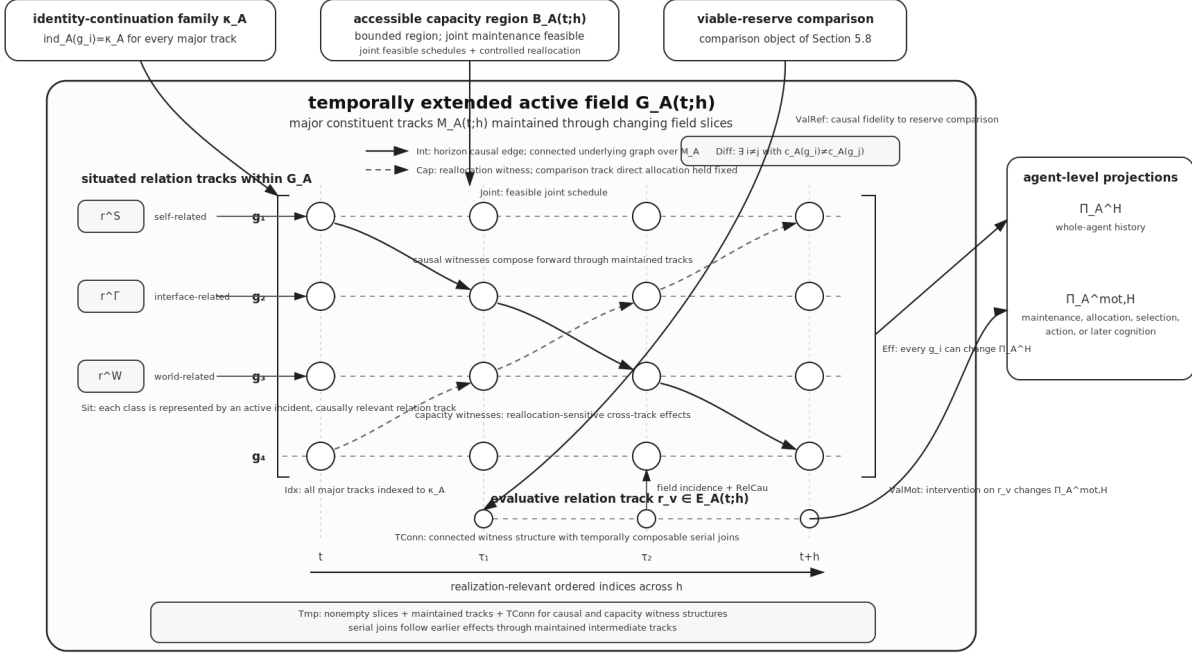


Figure 2: Temporally extended active-field schema showing maintained major constituent tracks, causal and capacity relations, and self-, interface-, world-related, and evaluative organization.

Strong connectivity is not required. The condition excludes a partition of the declared major constituent tracks with no intervention-sensitive causal relation crossing it.

Integration among field constituents is supplemented by **agent-scale continuation efficacy**. Every major track must be capable, under some admissible intervention, of changing a declared whole-agent history projection $\Pi_A^{\mathcal{H}}$:

$$\text{Eff}_{A,\chi}(t) \iff \forall g_i \in M_A(t;h) \exists \iota_i \in \mathcal{I}_i : \Pi_A^{\mathcal{H}}[\mathcal{H}_A^{\iota_i}] \neq \Pi_A^{\mathcal{H}}[\mathcal{H}_A].$$

This prevents a purely local overwritten variable from counting as agent-scale field efficacy.

4.3 Shared finite-capacity coupling

Finite capacity enters the construction at two related levels. C_6 requires candidate organizations that are separately feasible while unrestricted joint maintenance is infeasible, introducing capacity-constrained selection. At the field stage, the organization actually maintained must remain feasible:

$$\text{Joint}_{A,\chi}(t) \iff \exists \sigma \in \Sigma_A(G_A; t, h) : \mathbf{D}_A(G_A, \sigma; t, h) \in \mathcal{B}_A(t; h).$$

Define an undirected capacity graph $\mathcal{G}_A^{\text{cap}} = (M_A, E_A^{\text{cap}})$. An edge $\{g_i, g_j\}$ exists when two capacity-feasible schedules under the same exogenous conditions and accessible capacity region differ only through an admissible change to the direct allocation of one constituent and its causal consequences, with the direct allocations of all other major constituents held fixed, and the realized history of another constituent changes. Then

$$\text{Cap}_{A,\chi}(t) \iff \text{Joint}_{A,\chi}(t) \wedge \text{Conn}(\mathcal{G}_A^{\text{cap}}(t; h)).$$

Thus **Cap** requires cross-constituent sensitivity to allocation within one bounded heterogeneous capacity organization; it does not require one scalar resource or pairwise competition for a single fungible resource pool.

4.4 Persistent self-indexing and situated organization

Let $\mathfrak{K}_A$ be the declared family of identity-continuation families that are causally admissible for the candidate organization; it may not be restricted merely to make the indexing condition true. Let $\text{ind}_A : M_A(t; h) \rightarrow \mathfrak{K}_A$ assign the family relative to which each major constituent's maintained memories, predictions, controls, and actions are organized. The assignment must be fixed by the realization-level causal organization and invariant under mere relabelling. Persistent self-indexing requires

$$\boxed{\text{Idx}_{A,\chi}(t) \iff \forall g_i \in M_A(t; h) : \text{ind}_A(g_i) = \kappa_A.}$$

A first-person label, account identifier, or session key does not establish this relation.

Let $\text{Rel}(G_A)$ be the active relation tracks and Inc_A their incidence relation with major constituents. For each active relation track r , let $\mathcal{J}_r$ be the admissible interventions on variables realizing that relation. Such interventions must respect the declared boundary and realization constraints and must not directly overwrite the major-track or whole-agent outcome variables used to establish their effects. Relation-level causal relevance is

$$\boxed{\text{RelCau}_{A,\chi}(r, t) \iff \exists \iota_r \in \mathcal{J}_r : \left[\Pi_A^{\mathcal{H}}[\mathcal{H}'_r] \neq \Pi_A^{\mathcal{H}}[\mathcal{H}_A] \right. \\ \left. \vee \exists g_j \in M_A(t; h) : \pi_j^{\mathcal{H}}[\mathcal{H}'_r] \neq \pi_j^{\mathcal{H}}[\mathcal{H}_A] \right].}$$

Situated organization requires at least one causally relevant self-related, interface-related, and world-related relation to be incident on the field:

$$\boxed{\text{Sit}_{A,\chi}(t) \iff \forall \alpha \in \{S, \Gamma, W\} \exists r, g_i : r \in \text{Rel}(G_A) \cap \Lambda_A^\alpha \wedge \text{Inc}_A(r, g_i) \wedge \text{RelCau}_{A,\chi}(r, t).}$$

The field is thereby organized relative to the agent's own boundary, capacities, and interventions rather than as an external scene model alone.

4.5 Viable reserve, field-integrated valence, and motivation

Agenthood requires identity-preserving continuation under the declared disturbance class. Field-integrated valence concerns whether differences in viable continuation become causal constituents of selection and field organization.

For each disturbance d , define the identity-preserving continuation set

$$\mathcal{R}_{A,\kappa}^d(x, t; h) = \{x' \in \mathcal{R}_A^d(x, t; h) \mid \kappa_A^{t,t+h}(x, x')\}.$$

The **viable reserve profile** is the disturbance-indexed family

$$\boxed{\mathcal{V}_A(x; t, h, \mathcal{D}_A) = \left(\mathcal{R}_{A,\kappa}^d(x, t; h) \right)_{d \in \mathcal{D}_A} .}$$

A realization may enrich the comparison with burdens, deadlines, or capacity margins only when these are represented in its state, demand, or feasibility relations. The reserve preorder must be grounded in declared dominance relations and may not be selected from an action label, externally assigned reward, or the outcome the analysis is intended to obtain. Under matched disturbances, componentwise improvement cannot count as worsening, and a strict classification requires at least one declared strict improvement or loss; otherwise profiles remain equivalent or incomparable. An actual-transition comparison is admissible only under a

declared correspondence between the pre- and post-transition profile domains, including the relevant horizons, disturbance classes, disturbance correspondence, represented burdens, deadlines, and capacity margins. If no such correspondence is justified, no actual-transition valence classification is assigned. Under an admissible cross-time comparison, an actual transition is positive when the post-transition profile strictly improves, neutral when the profiles are equivalent, negative when it strictly worsens, and mixed when the profiles are incomparable under the declared preorder.

Prospective comparisons contrast reserve profiles following candidate actualization with profiles following a matched alternative completion or non-actualization under the same declared exogenous conditions, disturbance correspondence, horizon, and represented constraints.

Let $\mathfrak{C}_A^{\text{val}}(t; h)$ be the declared class of represented actual or prospective viable-reserve comparisons. For $\nu \in \mathfrak{C}_A^{\text{val}}(t; h)$, let $\text{Cmp}_A(\nu)$ denote the compared reserve profiles and their declared comparison structure, and let $\sim_\nu$ be the realization-grounded equivalence relation identifying reserve comparisons with the same evaluative significance for that represented relation. Finer reserve differences within one $\sim_\nu$ class need not be distinguished. Let $\mathcal{J}_A^{\text{res}}(\nu)$ be admissible interventions on variables determining ν that do not directly overwrite the evaluative relation or later motivational outcomes, and let

$$\bar{\mathcal{J}}_A^{\text{res}}(\nu) := \{\text{id}\} \cup \mathcal{J}_A^{\text{res}}(\nu)$$

include the unmodified baseline. For an active relation track r , define evaluative fidelity by

$$\begin{aligned} \text{Fid}_{A,\chi}(r, \nu, t) \iff & \left[\exists \iota_1, \iota_2 \in \bar{\mathcal{J}}_A^{\text{res}}(\nu) : \text{Cmp}_A^{\iota_1}(\nu) \sim_\nu \text{Cmp}_A^{\iota_2}(\nu) \right] \\ & \wedge \forall \iota_1, \iota_2 \in \bar{\mathcal{J}}_A^{\text{res}}(\nu) : \\ & \quad \text{Cmp}_A^{\iota_1}(\nu) \sim_\nu \text{Cmp}_A^{\iota_2}(\nu) \\ & \implies \pi_r^{\mathcal{H}}[\mathcal{H}_A^{\iota_1}] \neq \pi_r^{\mathcal{H}}[\mathcal{H}_A^{\iota_2}]. \end{aligned}$$

Thus fidelity requires at least one nontrivial evaluative reserve contrast and preservation of every evaluatively relevant distinction across the declared causal contrasts. Let $\mathcal{E}_A(t; h)$ contain exactly those active relation tracks r for which there exists $\nu \in \mathfrak{C}_A^{\text{val}}(t; h)$ satisfying $\text{Fid}_{A,\chi}(r, \nu, t)$ and at least one admissible reserve intervention that changes both the represented comparison and r without directly overwriting either r or the later motivational outcomes.

Let $\Pi_A^{\text{mot}, \mathcal{H}}$ project histories onto the declared motivational outcomes, including field maintenance, capacity allocation, candidate activation or actualization, outward action, or later candidate availability. An intervention on an evaluative relation is admissible for this test only when it changes that relation without directly overwriting those motivational outcomes. Field-integrated valence and motivation require at least one evaluative relation to be incident on a major constituent, causally relevant to the field, and capable of altering that projection:

$$\text{ValMot}_{A,\chi}(t) \iff \exists r_v \in \mathcal{E}_A(t; h) \exists g_i \in M_A(t; h) \exists \iota_v \in \mathcal{J}_{r_v} : \text{Inc}_A(r_v, g_i) \wedge \text{RelCau}_{A,\chi}(r_v, t) \wedge \Pi_A^{\text{mot}, \mathcal{H}}[\mathcal{H}_A^{\iota_v}] \neq \Pi_A^{\text{mot}, \mathcal{H}}[\mathcal{H}_A].$$

Neutral contents, conflict among incomparable reserve profiles, risk, sacrifice, and motivational impairment remain compatible with the condition. The predicate requires viability significance to be part of the active field's causal organization; it does not impose continuous positive or negative valence or a scalar reward function.

4.6 Temporal unity and conjunctive definition

Temporal unity is evaluated on the time-extended field rather than on one simultaneous slice. Major constituent tracks extend across the horizon while causal and capacity relations may occur at different times. For $\alpha \in \{\text{cau}, \text{cap}\}$, let a temporal witness $w = (g_i, g_j, \tau^-, \tau^+)$ be supplied by the same admissible causal or controlled-reallocation contrast used to establish the corresponding horizon-level edge, beginning at τ^-

and producing a later realized difference on g_j at τ^+ , with $t \leq \tau^- < \tau^+ \leq t + h$. A witness set is **time-respecting connected** when its underlying undirected track graph is connected and every serial pair w, w' with $\text{tar}(w) = \text{src}(w')$ satisfies

$$\tau_w^+ \leq \tau_{w'}^- \quad \text{and the difference produced by } w \text{ remains present on the intermediate track at } \tau_{w'}^-.$$

The matched baseline and modified source organizations used by w' must also be realization-compatible with the corresponding baseline and modified organizations supplied by w at that intermediate track. This condition permits common-source and common-target branches without imposing a total temporal order; it requires causal order and source compatibility only where one witness supplies the source organization of a later witness.

$\text{Tmp}_{A,\chi}(t)$ is the conjunction of a nonempty field slice at each realization-relevant index, maintenance of the declared major tracks, time-respecting connected witness structures for both causal and capacity relations, and persistence of identity indexing, situated incidence, and agent-scale efficacy. It additionally requires at least one witness evaluative track r_v for $\text{ValMot}_{A,\chi}(t)$ whose track continuity, field incidence, and causal fidelity to the same represented viable-reserve comparison are maintained through at least two causally dependent transitions. The classification is relative to the declared scale and horizon; sensitivity across plausible nearby horizons must be reported in realization assessment.

Definition 8 (structural consciousness).

$$\begin{aligned} \text{SC}_\chi(A, t) \iff & \text{Cognitive}_\chi(A, t) \wedge \text{Diff}_{A,\chi}(t) \wedge \text{Int}_{A,\chi}(t) \wedge \text{Eff}_{A,\chi}(t) \\ & \wedge \text{Cap}_{A,\chi}(t) \wedge \text{Idx}_{A,\chi}(t) \wedge \text{Sit}_{A,\chi}(t) \\ & \wedge \text{ValMot}_{A,\chi}(t) \wedge \text{Tmp}_{A,\chi}(t). \end{aligned}$$

Definition 8 is conjunctive. The eight field predicates form a dependency-structured conjunction; in particular, Tmp specifies temporal maintenance and composition of causal, capacity, identity-indexed, situated, agent-effective, and evaluative relations introduced by the other field conditions. Because Definition 5 includes Definition 4, all five agenthood conditions, all seven cognition conditions, and all eight field conditions are required at the declared context. The hierarchy is non-compensatory: satisfaction of one relation does not substitute for absence of another. A homogeneous controller fails differentiation; disconnected differentiated processes fail field-wide causal integration; a locally coupled subsystem with no whole-agent effect fails efficacy; independent resource use fails the connected capacity condition; a coordinated plurality without one identity family fails persistent self-indexing; an external scene model without causally active self and interface relations fails situated organization; a field whose viable-reserve differences never alter maintenance, selection, allocation, action, or later cognition fails field-integrated valence; and temporally incompatible fragments fail temporal unity.

5. Structural and phenomenological consequences

5.1 Unity and agent-relative organization

Define structural unity by connected causal integration, connected capacity coupling, one identity-continuation family with situated self-interface-world organization, and agent-scale efficacy over the same horizon. By Definition 8,

$$\text{SC}_\chi(A, t) \implies \text{Unity}_\chi(G_A(t; h), t).$$

Differentiation supplies distinct contents; causal integration and capacity coupling connect the declared major constituents; identity indexing and situated organization relate them to one continuing agent; agent-scale efficacy relates their changes to that agent's future possibilities; and temporal unity maintains the organization through causally dependent transitions.

The identity-continuation family introduced at the agenthood stage remains the reference organization through the later construction. Historical deformation concerns that agent’s present organization; active contents are indexed and situated relative to its continuation and boundary; efficacy concerns its later possibilities; viable-reserve comparisons concern those same continuations; and temporal unity maintains these relations through causally dependent transitions. Perspective is therefore agent-relative without requiring a separate reflective self-representation.

The field is unified without being homogeneous. Universal broadcast is not required: structural availability concerns influence on the relevant agent-level organization over the stated horizon. Attention can change field membership or precision through limited-capacity allocation but is not identical to field-wide causal integration.[9][10][17][18][25]

5.2 Qualitative occurrence, recurrence, and reinstatement

For a system satisfying Definition 8 under an admissible realization interpretation, a realized differentiated occurrence in the declared qualitative comparison domain is a **qualitative occurrence** when it is a constituent of the complete field. Prior structural information concerning a qualitative relation and realization of the corresponding occurrence within that agent’s field are distinct organizational states. A first occurrence may be novel relative to every retained type while occupying realization-specific proximity relations to prior qualities; retained effects of that occurrence can contribute to historical deformation, later stabilization of a recurrent qualitative type, and partial reinstatement on subsequent encounters. A learned type, name, autobiographical association, or valence need not precede the occurrence.

The same retained type may occur within different larger composites while shape, texture, action significance, autobiographical relation, and valence differ. The recurrent qualitative relation is therefore not identical to any one object, word, autobiographical association, or affective association. Present phenomenological organization includes both that recurrent relation and the historical, bodily, evaluative, and action-related relations currently integrated with it. The formalism does not derive one universal qualitative geometry; each realization must justify its content identity, recurrence, and proximity relations. Partial relational reinstatement permits a present cue to realize again a declared nonempty part of earlier sensory, bodily, world-related, or evaluative organization without exact state replay.

These relations can operate within one episode. A differentiated bodily or perceptual occurrence can be situated relative to the agent’s interface and a world-related cause, recruit organization retained from prior interaction, alter prospective viable-reserve relations, and change which competing contents or actions remain maintainable. Selection under finite capacity then changes the organization from which subsequent contents and actions become available. The episode illustrates joint realization of qualitative, historical, situated, evaluative, capacity-sensitive, and temporal relations within one continuing field.

5.3 Valenced phenomenological organization

Viable-reserve comparison is built from the identity, continuation, disturbance, capacity, and field relations already defined. When an active evaluative relation faithfully represents an actual or prospective reserve difference and causally affects maintenance, allocation, selection, action, or later cognition, viability significance is integrated into the field through ValMot.

Within SC, **valenced feeling** is the phenomenological organization in which that viability significance is a realized, self-indexed, situated, causally active field constituent. Neutral contents, mixed or incomparable reserve profiles, conflicting motivations, and impaired motivation remain compatible with the condition; protective regulation outside the field is not thereby classified as conscious feeling.

6. Private first-person access and relation to existing theories

6.1 External description and first-person standpoint

Let O be an external observer and A an agent realizing a subjective field. Increasingly complete third-person information about A can identify its states, causal relations, histories, and field organization without making

those occurrences constituents of O 's own field. Let $\text{Desc}_O(A, t)$ denote such an external structural description and $\text{FPAcc}_O(A, t)$ direct access to those occurrences from A 's first-person standpoint. SC contains no rule by which $\text{Desc}_O(A, t)$ becomes $\text{FPAcc}_O(A, t)$. External structural description therefore does not, within the account, establish occupation of the other agent's first-person standpoint.

The distinction is between two epistemic relations to one physical realization: external description of the organization and first-person occurrence as a constituent of the realizing agent's field. It does not require a nonphysical substance or an additional causal variable. This is an epistemic distinction in the premises, not an additional constitutive condition.

The distinction also does not insulate realization claims from empirical assessment. Whether the predicates of Definitions 4, 5, and 8 are physically realized is a third-person causal question. Section 9 specifies the evidence required for those claims. What remains unavailable to the observer *as observer* is occupation of the other system's first-person standpoint.

6.2 Relation to neighbouring consciousness theories

SC shares explanatory targets with several existing theories while proposing a different constitutive organization. Block's distinction between access and phenomenal consciousness and Chalmers's distinction between functional explanation and experience separate targets that SC does not reduce to behavioural report.[2][3]

IIT and SC both assign constitutive significance to physical causal organization, but specify different constitutive structures and routes of construction. IIT proceeds from phenomenological axioms to intrinsic cause-effect structure and associated measures; SC proceeds through continuing agenthood, history-sensitive cognition, and the field conjunction of Definition 8.[15][16] SC's Int is intervention-sensitive connectivity over declared major tracks, supplemented by agent-scale efficacy and a separate capacity relation.

Global Neuronal Workspace accounts place broad availability or broadcast at the centre of conscious access.[10][17][18] SC permits broad availability but does not make universal broadcast constitutive. Recurrent Processing Theory emphasizes recurrent neural interaction,[57] while higher-order approaches make representation of lower-order states central.[58] Either relation may occur within SC when it realizes the declared predicates, but neither is introduced as a separate sufficient condition for the complete field.

Structuralist and phenomenal-unity approaches provide close comparisons for qualitative relations and belonging-together.[40][41][45][46] SC places realization-grounded qualitative differentiation within the larger organization of one continuing agent's integrated, capacity-sensitive, situated, historically formed, and evaluatively organized field. The comparison concerns explanatory commitments; SC's claim is the particular conjunction defined here.

7. Biological realization: candidate mappings and empirical constraints

Substrate independence applies to the definition, not to physical implementation. Biological realization must map neural, bodily, metabolic, sensory, motor, and regulatory processes to the same causal roles without assuming one-to-one anatomical localization.

7.1 Capacity, embodiment, and retained history

For a human candidate H , $\mathcal{B}_H(t; h)$ may be realized through heterogeneous limits involving metabolic supply, ionic restoration, supportable activation, communication bandwidth, regulatory margin, and causal deadlines.[24] Working memory and attention provide candidate allocation relations because limited maintenance and nonuniform allocation alter present availability.[9][18][25] C_6 concerns separately feasible but jointly incompatible candidates; Cap additionally requires controlled reallocation through one major constituent to alter another while the latter's direct allocation is held fixed.

Situated organization may include bodily condition and capabilities, sensory and motor exchange, autonomic transformation paths, and world regularities that persist partly independently of control. Interoceptive and homeostatic signalling provide candidate self-related and evaluative relations,[11][12] while tool use,

prostheses, and body-ownership manipulations show why the self-interface-world classes are causal rather than simply anatomical.[21][22][23][26][27]

Learning, perceptual history, memory, and cue-dependent reinstatement provide candidate mechanisms for C_1 – C_5 ; competition and prioritization for C_6 ; and selection-dependent changes in later availability for C_7 . [13][28]

7.2 Field and non-field processing; neurological constraints

A process can alter later field organization without itself satisfying the field conditions. Evidence for non-conscious influence on later processing is compatible with that distinction.[14]

Recent sensory history alters perception; partial cues reinstate distributed hippocampal and cortical organization; higher-order thalamic activity can regulate history-dependent updating; priority alters working-memory resource allocation; bodily self-consciousness depends on interacting multisensory and self-attribution processes; and anaesthesia changes large-scale integration and temporal network dynamics.[50][51][52][53][54][55][56] These are candidate realization relations, not one-to-one proofs of SC predicates.

Comparative findings in nematodes, bees, and cephalopods further constrain assumptions about scale, boundary, and anatomical centralization.[60][61][62][63][64] Positive biological classification requires evidence for the complete conjunction within one candidate organization and declared context.

8. Artificial realization: architectural screening and candidate mappings

Artificial systems can combine predictive models with persistent state, memory, tools, planning loops, world models, and orchestration. These features make agent-level organization an empirical property of a deployed system rather than something inferable from model scale or self-report.[4][5][6][7][8]

A base language or multimodal model may contain differentiated mutually conditioning representations and generate reports of memory, intention, pain, uncertainty, or self-reflection.[32][33] Such capacities do not by themselves establish one persistent agent, identity-continuation family, effective boundary, viability relation, or ongoing action loop.

Stateful agent architectures add candidate mechanisms: persistent memory can make earlier interaction causally effective; tools can provide outward efficacy; repeated perception-action loops can support extended regulation; world models can contribute world-related organization; and scheduling or resource limits can realize capacity relations. Each counts only through the causal role it actually realizes. A sequence of services does not by itself establish one field-wide causal or capacity organization.

Multi-agent coordination likewise need not constitute unity, while a distributed implementation can count as one candidate when the relevant identity, boundary, causal, capacity, situated, and temporal relations occur at that scale. Model weights, context length, storage, retrieval, tools, planning, reward mechanisms, world models, and runtime duration are therefore candidate realization mechanisms, not scalar indicators of structural consciousness.[34][35][36]

9. Empirical identification

The formal predicates state causal properties of a physical candidate. Their assessment therefore requires realization-specific evidence rather than behavioural resemblance alone. Four evidential states are useful for each predicate: **demonstrated satisfaction**, **demonstrated failure**, **unresolved evidence**, and **inaccessibility**. Because Definition 8 is conjunctive, demonstrated failure of any required predicate establishes non-membership at the selected context; unresolved or inaccessible predicates prevent a positive classification unless another required predicate is already demonstrated absent.

9.1 Realization assessment

Assessment specifies the candidate system, boundary, scale s , horizon h , disturbance class $\mathcal{D}_A$, accessible variables, and admissible realization interpretation ρ_Y . Boundary variables, major tracks, intervention classes, capacity objects, identity families, comparison maps, and reserve rules must be justified independently of whether they make the candidate satisfy a predicate. Definition 4 is assessed before Definition 5, and positive progression to Definition 8 requires demonstrated cognition.

Predicate evidence then uses matched causal contrasts. Prior exposure and cue history test retained deformation, composite formation, recruitment, and reinstatement; a C_2 contrast must change later non-directly-assigned organization rather than merely preserve the imposed retained value. Internal and external interventions test boundary coupling and agent-scale efficacy. Controlled reallocations test **Cap** while matching other direct allocations and independently manipulable schedule variables. Variations in bodily, environmental, interface, or action contingencies test situated organization. Reserve interventions test whether evaluatively distinct comparisons remain distinguishable in an active relation and causally affect maintenance, allocation, selection, action, or later cognition. Temporal witnesses test whether causal and capacity relations compose through the horizon with maintained intermediate effects and compatible source organization.

Reports concerning memory, qualitative content, valence, intention, or conflict count as structurally grounded only when causal variation in the proposed underlying relation predicts corresponding changes in the report and relevant later control. Null outcomes are predicate-specific: local-only effects count against **Eff**, label-only content differences against realization-grounded **Diff**, absence of matched reallocation effects against **Cap**, failure of cue-dependent historical recruitment against reinstatement, and temporally incompatible witness structures against **Tmp**. Failure to establish a nontrivial evaluative reserve contrast or its faithful representation counts against the evaluative relation; where such a relation is established but fails to alter the declared motivational projection, the failure bears on **ValMot**.

9.2 Realization assessment and evaluation of the constitutive account

Realization assessment asks whether, given ρ_Y and χ , the candidate satisfies Definition 8. Demonstrated satisfaction establishes structural-consciousness classification at that context; failure of any required predicate establishes non-membership.

Evaluation of the constitutive account is a separate evidential question. Comparative evidence across conscious, unconscious, clinical, dissociated, biological, comparative, and artificial cases can test whether the proposed relations are preserved, disrupted, or dissociated with the target experience. Target-status evidence used in that comparison must be motivated independently of SC classification.[49] Realization studies therefore establish whether the proposed organization is present; comparative work evaluates whether that organization tracks the target phenomenon.

The constitutive necessity of a required predicate is evaluated by matched comparative cases. Evidence against necessity is provided by a case in which the predicate is absent while the remaining relevant organization and independently warranted evidence for the target subjective experience are preserved. Evidence for the proposed constitutive role is strengthened when selective disruption of the predicate’s realized relation produces the corresponding predicted alteration of the target organization under matched comparison conditions. The conjunctive classification establishes the formal consequence of predicate failure within SC; comparative dissociation and selective disruption evaluate the corresponding empirical necessity claim.

SC introduces no scalar consciousness measure. Any such measure would require separately validated observables, thresholds, weights, error models, and an aggregation rule.

10. Scope and conclusion

SC targets **unified, structured, agent-relative subjective experience from a continuing point of view**. Putative minimally structured or non-agentic forms discussed elsewhere are not classified by this construction.[42] Every classification is relative to a declared context and admissible realization interpretation;

substrate independence applies to the definition, while each realization remains constrained by its material processes, timing, resources, sensory transformations, historical formation, and failure modes.

The construction identifies one continuing agent as the bearer whose retained history shapes present cognition, whose differentiated contents participate in one causal and capacity-sensitive field, whose self-interface-world relations situate those contents, and whose feasible continuations give evaluative significance to present organization. Temporal unity maintains these relations through successive transitions, so present activity changes the organization from which later contents and actions become available. Complete physical realization of this conjunction is the proposed constitutive organization of the target subjective field.

Realization-specific evidence must establish the complete conjunction; biological and artificial mechanisms enter through the relations they realize rather than through anatomical similarity, model scale, memory size, tools, or report alone. The identity-continuation relation remains a stipulated realization-level criterion, with singular actual lineage an explicit identity hypothesis rather than a consequence of Definition 8 or general physics. External structural description and the agent’s first-person standpoint remain distinct epistemic relations to the realized organization.

No new empirical dataset is reported. Section 9 specifies realization assessment and comparative evaluation of the constitutive account.

AI-assistance disclosure

The research questions, conceptual framework, qualitative and epistemic reasoning, and principal theoretical arguments originated with the author. Generative AI systems were used during development for drafting assistance, mathematical formalization, literature-search support, consistency checking, and editorial revision. The author reviewed and revised the AI-assisted material and assumes full responsibility for the content of the manuscript.

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References

1. Nagel, T. (1974). “What Is It Like to Be a Bat?” *The Philosophical Review*, 83(4), 435–450. <https://doi.org/10.2307/2183914>
2. Block, N. (1995). “On a Confusion about a Function of Consciousness.” *Behavioral and Brain Sciences*, 18(2), 227–247. <https://doi.org/10.1017/S0140525X00038188>
3. Chalmers, D. J. (1995). “Facing Up to the Problem of Consciousness.” *Journal of Consciousness Studies*, 2(3), 200–219. <https://consc.net/papers/facing.html>
4. Chalmers, D. J. (2023). “Could a Large Language Model Be Conscious?” arXiv:2303.07103. <https://doi.org/10.48550/arXiv.2303.07103>

5. Butlin, P., Long, R., Bayne, T., Bengio, Y., Birch, J., Chalmers, D., Constant, A., Deane, G., Elmoznino, E., Fleming, S. M., Ji, X., Kanai, R., Klein, C., Lindsay, G., Michel, M., Mudrik, L., Peters, M. A. K., Schwitzgebel, E., Simon, J., & VanRullen, R. (2026). "Identifying Indicators of Consciousness in AI Systems." *Trends in Cognitive Sciences*, 30(6), 488–501. <https://doi.org/10.1016/j.tics.2025.10.011>
6. OpenAI. "OpenAI Agents SDK: Agents and Sessions." Accessed 30 July 2026. <https://openai.github.io/openai-agents-python/>
7. LangChain. "LangGraph Documentation: Persistence and Memory." Accessed 30 July 2026. <https://docs.langchain.com/oss/python/langgraph/persistence>
8. Microsoft. "Microsoft Agent Framework Overview." Accessed 30 July 2026. <https://learn.microsoft.com/en-us/agent-framework/overview/>
9. Baddeley, A. (2012). "Working Memory: Theories, Models, and Controversies." *Annual Review of Psychology*, 63, 1–29. <https://doi.org/10.1146/annurev-psych-120710-100422>
10. Dehaene, S., & Changeux, J.-P. (2011). "Experimental and Theoretical Approaches to Conscious Processing." *Neuron*, 70(2), 200–227. <https://doi.org/10.1016/j.neuron.2011.03.018>
11. Craig, A. D. (2002). "How Do You Feel? Interoception: The Sense of the Physiological Condition of the Body." *Nature Reviews Neuroscience*, 3, 655–666. <https://doi.org/10.1038/nrn894>
12. Damasio, A., & Carvalho, G. B. (2013). "The Nature of Feelings: Evolutionary and Neurobiological Origins." *Nature Reviews Neuroscience*, 14, 143–152. <https://doi.org/10.1038/nrn3403>
13. Squire, L. R., & Wixted, J. T. (2011). "The Cognitive Neuroscience of Human Memory Since H.M." *Annual Review of Neuroscience*, 34, 259–288. <https://doi.org/10.1146/annurev-neuro-061010-113720>
14. Kouider, S., & Dehaene, S. (2007). "Levels of Processing During Non-Conscious Perception: A Critical Review of Visual Masking." *Philosophical Transactions of the Royal Society B*, 362(1481), 857–875. <https://doi.org/10.1098/rstb.2007.2093>
15. Oizumi, M., Albantakis, L., & Tononi, G. (2014). "From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0." *PLOS Computational Biology*, 10(5), e1003588. <https://doi.org/10.1371/journal.pcbi.1003588>
16. Albantakis, L., Barbosa, L., Findlay, G., Grasso, M., Haun, A. M., Marshall, W., et al. (2023). "Integrated Information Theory (IIT) 4.0: Formulating the Properties of Phenomenal Existence in Physical Terms." *PLOS Computational Biology*, 19(10), e1011465. <https://doi.org/10.1371/journal.pcbi.1011465>
17. Dehaene, S., Kerszberg, M., & Changeux, J.-P. (1998). "A Neuronal Model of a Global Workspace in Effortful Cognitive Tasks." *Proceedings of the National Academy of Sciences*, 95(24), 14529–14534. <https://doi.org/10.1073/pnas.95.24.14529>
18. Mashour, G. A., Roelfsema, P., Changeux, J.-P., & Dehaene, S. (2020). "Conscious Processing and the Global Neuronal Workspace Hypothesis." *Neuron*, 105(5), 776–798. <https://doi.org/10.1016/j.neuron.2020.01.026>
19. Barandiaran, X. E., Di Paolo, E., & Rohde, M. (2009). "Defining Agency: Individuality, Normativity, Asymmetry, and Spatio-temporality in Action." *Adaptive Behavior*, 17(5), 367–386. <https://doi.org/10.1177/1059712309343819>
20. Di Paolo, E. A. (2005). "Autopoiesis, Adaptivity, Teleology, Agency." *Phenomenology and the Cognitive Sciences*, 4(4), 429–452. <https://doi.org/10.1007/s11097-005-9002-y>
21. Clark, A., & Chalmers, D. (1998). "The Extended Mind." *Analysis*, 58(1), 7–19. <https://doi.org/10.1093/analys/58.1.7>

22. Iriki, A., Tanaka, M., & Iwamura, Y. (1996). “Coding of Modified Body Schema During Tool Use by Macaque Postcentral Neurones.” *NeuroReport*, 7(14), 2325–2330. <https://doi.org/10.1097/00001756-199610020-00010>
23. Cardinali, L., Frassinetti, F., Brozzoli, C., Urquizar, C., Roy, A. C., & Farnè, A. (2009). “Tool-Use Induces Morphological Updating of the Body Schema.” *Current Biology*, 19(12), R478–R479. <https://doi.org/10.1016/j.cub.2009.05.009>
24. Attwell, D., & Laughlin, S. B. (2001). “An Energy Budget for Signaling in the Grey Matter of the Brain.” *Journal of Cerebral Blood Flow & Metabolism*, 21(10), 1133–1145. <https://doi.org/10.1097/0004647-200110000-00001>
25. Cowan, N. (2001). “The Magical Number 4 in Short-Term Memory: A Reconsideration of Mental Storage Capacity.” *Behavioral and Brain Sciences*, 24(1), 87–114. <https://doi.org/10.1017/S0140525X01003922>
26. Blanke, O., Slater, M., & Serino, A. (2015). “Behavioral, Neural, and Computational Principles of Bodily Self-Consciousness.” *Neuron*, 88(1), 145–166. <https://doi.org/10.1016/j.neuron.2015.09.029>
27. Tsakiris, M. (2010). “My Body in the Brain: A Neurocognitive Model of Body-Ownership.” *Neuropsychologia*, 48(3), 703–712. <https://doi.org/10.1016/j.neuropsychologia.2009.09.034>
28. Conway, M. A., & Pleydell-Pearce, C. W. (2000). “The Construction of Autobiographical Memories in the Self-Memory System.” *Psychological Review*, 107(2), 261–288. <https://doi.org/10.1037/0033-295X.107.2.261>
29. Shoemaker, S. S. (1959). “Personal Identity and Memory.” *The Journal of Philosophy*, 56(22), 868–902. <https://doi.org/10.2307/2022317>
30. Lewis, D. (1976). “Survival and Identity.” In A. O. Rorty (Ed.), *The Identities of Persons* (pp. 17–40). University of California Press. <https://doi.org/10.1525/9780520353060-002>
31. Parfit, D. (1984). *Reasons and Persons*. Clarendon Press.
32. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). “Attention Is All You Need.” *Advances in Neural Information Processing Systems*, 30. <https://doi.org/10.48550/arXiv.1706.03762>
33. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., et al. (2020). “Language Models Are Few-Shot Learners.” *Advances in Neural Information Processing Systems*, 33, 1877–1901. <https://doi.org/10.48550/arXiv.2005.14165>
34. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). “ReAct: Synergizing Reasoning and Acting in Language Models.” *International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.2210.03629>
35. Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). “Generative Agents: Interactive Simulacra of Human Behavior.” In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (Article 2, pp. 1–22). <https://doi.org/10.1145/3586183.3606763>
36. Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., & Yao, S. (2023). “Reflexion: Language Agents with Verbal Reinforcement Learning.” *Advances in Neural Information Processing Systems*, 36. <https://doi.org/10.48550/arXiv.2303.11366>
37. Norman, D. A., & Bobrow, D. G. (1975). “On Data-Limited and Resource-Limited Processes.” *Cognitive Psychology*, 7(1), 44–64. [https://doi.org/10.1016/0010-0285\(75\)90004-3](https://doi.org/10.1016/0010-0285(75)90004-3)
38. Seth, A. K., & Bayne, T. (2022). “Theories of Consciousness.” *Nature Reviews Neuroscience*, 23(7), 439–452. <https://doi.org/10.1038/s41583-022-00587-4>

39. Mudrik, L., Boly, M., Dehaene, S., Fleming, S. M., Lamme, V., Seth, A., & Melloni, L. (2025). "Unpacking the Complexities of Consciousness: Theories and Reflections." *Neuroscience & Biobehavioral Reviews*, 170, 106053. <https://doi.org/10.1016/j.neubiorev.2025.106053>
40. Bayne, T., & Chalmers, D. J. (2003). "What Is the Unity of Consciousness?" In A. Cleeremans (Ed.), *The Unity of Consciousness: Binding, Integration, and Dissociation* (pp. 23–58). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198508571.003.0002>
41. Bayne, T. (2010). *The Unity of Consciousness*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199215386.001.0001>
42. Wiese, W. (2020). "The Science of Consciousness Does Not Need Another Theory, It Needs a Minimal Unifying Model." *Neuroscience of Consciousness*, 2020(1), niaa013. <https://doi.org/10.1093/nc/niaa013>
43. Signorelli, C. M., Szczotka, J., & Prentner, R. (2021). "Explanatory Profiles of Models of Consciousness - Towards a Systematic Classification." *Neuroscience of Consciousness*, 2021(2), niab021. <https://doi.org/10.1093/nc/niab021>
44. Niikawa, T., Miyahara, K., Hamada, H. T., & Nishida, S. (2022). "Functions of Consciousness: Conceptual Clarification." *Neuroscience of Consciousness*, 2022(1), niac006. <https://doi.org/10.1093/nc/niac006>
45. Lyre, H. (2022). "Neurophenomenal Structuralism. A Philosophical Agenda for a Structuralist Neuroscience of Consciousness." *Neuroscience of Consciousness*, 2022(1), niac012. <https://doi.org/10.1093/nc/niac012>
46. Kob, L. (2023). "Exploring the Role of Structuralist Methodology in the Neuroscience of Consciousness: A Defense and Analysis." *Neuroscience of Consciousness*, 2023(1), niad011. <https://doi.org/10.1093/nc/niad011>
47. Schurger, A., & Graziano, M. (2022). "Consciousness Explained or Described?" *Neuroscience of Consciousness*, 2022(1), niac001. <https://doi.org/10.1093/nc/niac001>
48. Ellia, F., Hendren, J., Grasso, M., Kozma, C., Mindt, G., Lang, J. P., Haun, A. M., Albantakis, L., Boly, M., & Tononi, G. (2021). "Consciousness and the Fallacy of Misplaced Objectivity." *Neuroscience of Consciousness*, 2021(2), niab032. <https://doi.org/10.1093/nc/niab032>
49. Negro, N. (2024). "(Dis)confirming Theories of Consciousness and Their Predictions: Towards a Lakatosian Consciousness Science." *Neuroscience of Consciousness*, 2024(1), niae012. <https://doi.org/10.1093/nc/naie012>
50. Cicchini, G. M., Mikellidou, K., & Burr, D. C. (2024). "Serial Dependence in Perception." *Annual Review of Psychology*, 75, 129–154. <https://doi.org/10.1146/annurev-psych-021523-104939>
51. Zheng, L., Gao, Z., McAvan, A. S., Isham, E. A., & Ekstrom, A. D. (2021). "Partially Overlapping Spatial Environments Trigger Reinstatement in Hippocampus and Schema Representations in Prefrontal Cortex." *Nature Communications*, 12, 6231. <https://doi.org/10.1038/s41467-021-26560-w>
52. Hosford, P. S., Mei, H., Tagomori, H., Hayde, C. P., Abdelaal, M. S., Tagomori, H., Nakajima, M., & Schmitt, L. I. (2026). "Control of Representation Updating by Higher-Order Thalamus Enables History-Based Decision-Making." *Neuron*, advance online publication. <https://doi.org/10.1016/j.neuron.2026.04.045>
53. Li, H.-H., Sprague, T. C., Yoo, A. H., Ma, W. J., & Curtis, C. E. (2025). "Neural Mechanisms of Resource Allocation in Working Memory." *Science Advances*, 11(15), eadr8015. <https://doi.org/10.1126/sciadv.adr8015>
54. Brunello, N., Diana, L., Sritharan, J., Glisic, M., Nef, T., Verma, R. K., & Zito, G. A. (2025). "A Systematic Review and Meta-Analysis on the Neural Correlates of Bodily Self-Consciousness." *Neuroscience & Biobehavioral Reviews*, 179, 106420. <https://doi.org/10.1016/j.neubiorev.2025.106420>

55. Jang, H., Mashour, G. A., Hudetz, A. G., & Huang, Z. (2024). “Measuring the Dynamic Balance of Integration and Segregation Underlying Consciousness, Anesthesia, and Sleep in Humans.” *Nature Communications*, 15, 9164. <https://doi.org/10.1038/s41467-024-53299-x>
56. Ensel, S., Uhrig, L., Ozkirli, A., Hoffner, G., Tasserie, J., Dehaene, S., Van De Ville, D., Jarraya, B., & Pirondini, E. (2024). “Transient Brain Activity Dynamics Discriminate Levels of Consciousness During Anesthesia.” *Communications Biology*, 7, 716. <https://doi.org/10.1038/s42003-024-06335-x>
57. Lamme, V. A. F. (2006). “Towards a True Neural Stance on Consciousness.” *Trends in Cognitive Sciences*, 10(11), 494–501. <https://doi.org/10.1016/j.tics.2006.09.001>
58. Rosenthal, D. M. (2005). *Consciousness and Mind*. Clarendon Press. <https://doi.org/10.1093/oso/9780198236979.001.0001>
59. Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/CBO9780511803161>
60. Randi, F., Sharma, A. K., Dvali, S., & Leifer, A. M. (2023). “Neural Signal Propagation Atlas of *Caenorhabditis elegans*.” *Nature*, 623, 406–414. <https://doi.org/10.1038/s41586-023-06683-4>
61. Giurfa, M., Zhang, S., Jenett, A., Menzel, R., & Srinivasan, M. V. (2001). “The Concepts of ‘Sameness’ and ‘Difference’ in an Insect.” *Nature*, 410(6831), 930–933. <https://doi.org/10.1038/35073582>
62. Solvi, C., Gutierrez Al-Khudhairy, S., & Chittka, L. (2020). “Bumble Bees Display Cross-Modal Object Recognition Between Visual and Tactile Senses.” *Science*, 367(6480), 910–912. <https://doi.org/10.1126/science.aay8064>
63. Kuuspalu, A., Cody, S., & Hale, M. E. (2022). “Multiple Nerve Cords Connect the Arms of Octopuses, Providing Alternative Paths for Inter-Arm Signaling.” *Current Biology*, 32(24), 5415–5421.e3. <https://doi.org/10.1016/j.cub.2022.11.007>
64. Olson, C. S., Schulz, N. G., & Ragsdale, C. W. (2025). “Neuronal Segmentation in Cephalopod Arms.” *Nature Communications*, 16, 443. <https://doi.org/10.1038/s41467-024-55475-5>